IND-SMOOTHNESS OF AN ALGEBRA IS STABLE UNDER INFINITESIMAL DEFORMATIONS

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Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- One-step lifting through a square-zero ideal. lemma-lift-once (carded L26805). Rewrites a finitely presented A as B/J with B smooth and J finitely generated inside IB.
- Killing a finitely generated ideal inside IB. lemma-lift-twice (carded L26806). Produces the smooth B' with B -> B' -> Lambda killing J.
- Colimit criterion for ind-smooth maps. Algebra, lemma when-colimit (cross-chapter, named). Frames the claim as factoring every finitely presented A -> Lambda through a smooth algebra.

Source reference. Original reference 07C3: focused-seeds/geometry/S9-smo-proposition-lift/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CK: sources/stacks-smoothing/pinned-original-source.tex, source lines 1455–1455. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CL: sources/stacks-smoothing/pinned-original-source.tex, source lines 1525–1525. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let $R \to \Lambda$ be a ring map. Let $I \subset R$ be an ideal. Assume that*

(1) *I is nilpotent,*
(2) *$\Lambda/I\Lambda$ is a filtered colimit of smooth R/I-algebras, and*
(3) *$R \to \Lambda$ is flat.*

Then Λ is a filtered colimit of smooth R-algebras.

Proof. Since $I^n = 0$ for some n, it follows by induction on n that it suffices to consider the case where $I^2 = 0$. Let $\varphi : A \to \Lambda$ be an R-algebra map with A of finite presentation over R. We have to find a factorization $A \to B \to \Lambda$ with B smooth over R, see Algebra, Lemma 07C3. By Lemma 07CK we may assume that

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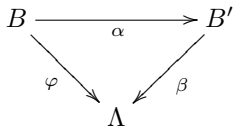
$A = B/J$ with B smooth over R and $J \subset IB$ a finitely generated ideal. By Lemma 07CL we can find a commutative diagram



of R-algebras with B' smooth over R such that $\alpha(J) = 0$. Thus α factors as $B \to A \to B'$ and the proof is complete. $\square$